



Idea Layer

Generate and refine research ideas.

Multi-Agent Discussion



Diverse perspectives

Propose Ideas



Generate candidates

Consensus Idea



Align and select



Planning Layer

Decompose ideas into concrete plans.

Plan



Tasks & milestones

Agent



Assign & planning

Refine Plan



Review & update

Finalize Plan



Ready to code

No

Good enough ?

Yes



Coding Layer

Implement, debug, and leverage tools.

Tools Call



Constructing context

Agent



Read & Understand

Coding



Test & update

Finalize Codes



Ready to execute

No

Bug free ?

Yes



Experiment Layer

Run experiments, and analyze results

Run on Server



Compute & storage

Results



Metric & logs

Iterate



Adjust & rerun



Writing Layer

Write, visualize, and polish the paper.

Outline



Structure paper

Figures



Visualize results

Draft



Write content

Review



Improve

Paper



Final manuscript